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A Review of Resistance Training for Adolescents: History, Controversies, and Practice Guidelines

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Abstract This study aims to systematically review the development history, related controversies, and practical guidance principles of resistance training for adolescents. The research shows that resistance training for adolescents has gradually shifted from being regarded as a "dangerous activity" in the early stages to a safe and effective training method supported by scientific evidence. The current controversies mainly focus on its safety and injury risks. The existing empirical evidence indicates that with effective supervision and scientific guidance, resistance training for adolescents not only has controllable safety but also significantly improves athletic performance, promotes physical development, and enhances mental health. Several professional associations have accordingly released training guidelines. However, the existing research and guidelines have not yet established a unified standard for "effective supervision", and generally ignore the confounding effects of age and gender. Future research should develop gender- and age-specific training plans, standardize supervision practices, and further promote resistance training in physical education.

Keywords Adolescents; Resistance Training; Safety; Supervision; Guideline Development.

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1. Introduction

Resistance training offers numerous benefits to the physical fitness of adolescents, but it is not widely practiced. Recent epidemiological data indicate that the level of physical activity among adolescents worldwide is far below the expected level, and problems such as obesity, sedentary behavior, and muscle atrophy are becoming increasingly prominent (Aira et al., 2021; Yan & Chen, 2024). Based on this background, intervention strategies aimed at optimizing the physical fitness and health of adolescents deserve attention. Resistance training not only significantly improves the muscle strength and overall athletic performance of adolescents (Granacher et al., 2016; Bagheri et al., 2023), but also has positive effects on the health of adolescents (Ribeiro et al., 2022; Kosmas, 2023). For example, advantages in terms of body composition and mental health, including reducing body fat, increasing bone density, improving the heart function of obese adolescents, and regulating insulin sensitivity in overweight adolescents. Although resistance training has many benefits, it is not widely practiced in the practice of adolescent physical education. The main reasons are safety concerns and outdated physiological concepts. In conclusion, this review emphasizes the significant public health impact of promoting safe and effective resistance training, especially in the context of the decline in physical activity levels among adolescents.

Resistance training is regarded as hindering bone growth and prone to injuries. These safety concerns and worries about bone development have labeled it as "unsafe". However, after long-term scientific verification, this misunderstanding has been overturned. Supervised and scientifically designed resistance training has significant positive effects on young people (Moran et al., 2021; Sortwell et al., 2022). Evidence shows that under supervision, the injury rate of adolescents participating in resistance training is much lower than that of regular sports activities such as football and basketball in physical education (Faigenbaum et al., 2020). As researchers and fitness associations deepen their understanding of resistance training, their professional consensus has changed, and guidelines for resistance training for adolescents have been issued. Associations such as NSCA, UKSCA, AAP, CSFE, and IOC have officially recognized the effectiveness and safety of resistance training for adolescents under scientific supervision (Behm et al., 2008; Lloyd et al., 2012; Lloyd et al., 2014; Bergeron et al., 2015; Lloyd et al., 2016; Stricker et al., 2020). The formation of this consensus indicates that it is necessary to establish a collaborative mechanism at the research, educational, and practical levels to enhance the understanding of coaches, parents, and adolescents about scientific resistance training, and through safe and efficient intervention strategies, promote the healthy development of adolescents' physical fitness. Therefore, this article aims to provide a comprehensive summary of the historical evolution of resistance training for adolescents and analyze the key points of controversy within the field. Additionally, it seeks to integrate current practice guidelines and educational insights, addressing long-standing misunderstandings and offering evidence-based recommendations for teachers, coaches, and policymakers.

2. Method

This study is a narrative review that aims to synthesize the history, controversy, and practice of resistance training in the existing literature. A comprehensive search was conducted in the SCOPUS, EBSCO, PubMed, and Google Scholar databases for relevant English-language articles published between 1980 and 2025. The search was guided by the following keywords: Adolescents, Resistance Training, Safety, and Supervision.

Inclusion criteria:

Target population of the study: Adolescents

Literature type: Journal article

Specific themes: resistance training; Supervise; Safety, etc.

Exclusion criteria:

Articles not relevant to the research question (e.g., not dealing with resistance training or adolescent populations)

Incomplete data in the article or unusable results (e.g., missing data, failure to report key variables)

Language does not meet the requirements (e.g., non-English articles)

3. Historical evolution

The historical evolution of resistance training for youth can be broadly divided into three developmental phases. In the early 1980s, resistance training was widely regarded as potentially harmful to the physical health and developmental processes of children and adolescents (Häkkinen et al., 1989). Prevailing misconceptions, coupled with limited empirical evidence, led to a cautious or even prohibitive stance toward its adoption in youth physical education and sports programs.

However, during the 1990s, as scientific inquiry deepened and empirical findings accumulated, a paradigm shift began to emerge (Figure 1). Research increasingly demonstrated that, when properly supervised and appropriately prescribed, resistance training could yield significant positive outcomes for young participants, including improvements in muscular strength, motor coordination, and overall physical fitness (Hamill, 1994).

Entering the late 20th and early 21st centuries, a substantial body of evidence consolidated the understanding that youth resistance training confers multiple physiological and psychological benefits (Faigenbaum et al., 2013). The consensus within the academic and professional communities gradually shifted from skepticism to advocacy. Leading organizations and professional associations began to formally endorse resistance training as a safe and beneficial component of youth physical development, emphasizing its role in promoting health, enhancing athletic performance, and fostering lifelong engagement in physical activity (McHenry & Nitka, 2022).

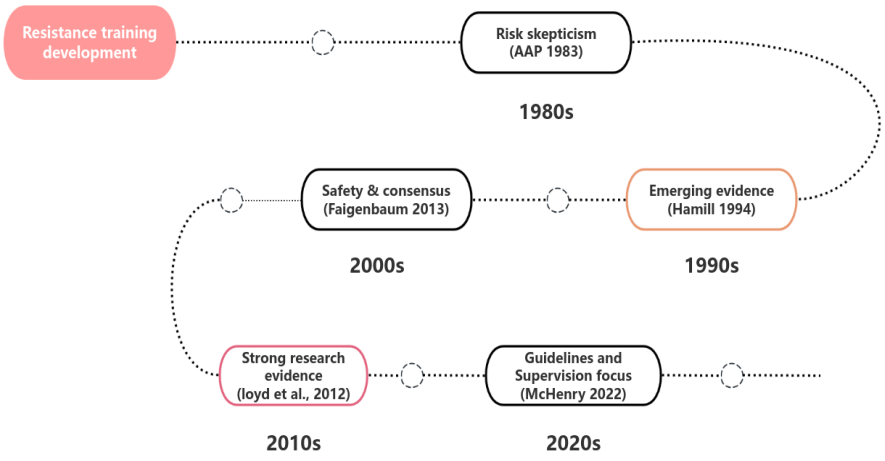

Figure 1: Timeline of Resistance Training Development

3.1 1987-1990s: From "Risk" to "Negative Impact"

Early studies and experts still raised doubts about the safety and effectiveness of resistance training for young people. The American Academy of Pediatrics (AAP) stated in a statement that due to the high injury rate, resistance training was dangerous for adolescents (American Academy of Pediatrics, 1983). Researchers expressed concerns about the potential risks that adolescents might face when participating in resistance training, including severe injuries (AAP, 1983) as well as adverse effects on the musculoskeletal, cardiovascular, and other physiological systems (Häkkinen et al., 1989). Although some institutions and scholars believed that resistance training was harmful to teenage athletes, the reported harmful results seemed to have selection bias. In the following years, more and more studies confirmed the safety and effectiveness of resistance training.

3.2 1990s - 2000s: Scientific evidence emerged

Resistance training for adolescents under supervision was both safe and effective. Hamill's 1994 study demonstrated that supervision by professionals could significantly reduce the injury rate in weightlifting and resistance training. Additionally, other studies have shown that children participating in scientific resistance training have a lower risk of joint sprains and muscle strains than adults (Myer et al., 2009). In conclusion, an increasing amount of evidence

indicated that after providing guidance on the correct use of weight training equipment and giving appropriate resistance training recommendations, the safety and effectiveness of resistance training for adolescents could be guaranteed (Lillegard et al., 1997; Tsolakis et al., 2004). With the emergence of scientific evidence, nowadays people have reached a consensus on the safety and effectiveness of resistance training and have promoted corresponding policies.

3.3 2000s - Present: Formation of Consensus and Promotion of Safety Education

With a deeper understanding of the scientific essence of resistance training for adolescents, the academic community and professional organizations gradually reached a broad consensus on its effectiveness and safety. Over the past three decades, empirical research has accumulated a large amount of evidence. It has fully demonstrated that resistance training, when implemented under scientific guidance and strict supervision, is both safe and highly effective for adolescents (Faigenbaum et al., 2013). The research results showed that systematic resistance training not only significantly enhances the muscle strength and neuromuscular coordination of adolescents, but also has positive effects on overall athletic performance and physical health - including reducing body fat percentage, increasing bone density, improving the heart function of obese adolescents, and enhancing insulin sensitivity in overweight individuals (Ribeiro et al., 2022; Kosmas, 2023). Supported by a series of evidence-based studies, the scientific value and health-promoting effects of resistance training have been increasingly recognized. Based on this, several authoritative international and domestic institutions and professional associations formulated safety norms and practice guidelines for adolescent resistance training, clearly stating that the safety and effectiveness of the training process should be guaranteed through a scientific teaching and supervision system (Lloyd et al., 2012; Lloyd et al., 2014; Bergeron et al., 2015; Lloyd et al., 2016; Stricker et al., 2020; McHenry & Nitka, 2022). At the same time, these organizations also emphasize the crucial role of parents, educators, coaches, and medical professionals in promoting youth sports activities, advocating that they should actively advocate and encourage adolescents to participate in resistance training to maximize the benefits in terms of physical health, athletic ability, and overall physical fitness development (McHenry & Nitka, 2022).

4. The Controversy Surrounding Resistance Training for Adolescents

For a long time, resistance training for adolescents has been in a controversial situation in academic discussions and public perception. The traditional perception of its dangers and high risk of injury has been deeply rooted in the minds of children, especially parents. However, now a lot of evidence has reshaped this traditional perception. Numerous studies have shown that scientific and supervised resistance training is safe and is an effective means for adolescents to enhance their physical fitness and prevent sports injuries (Chaabene et al., 2020; Tumkur Anil Kumar et al., 2021).

4.1 Early Controversy: Considerations Regarding Safety Risks

Early resistance training was considered unsafe and associated with a high risk of sports injuries. The probability of injuries for adolescents participating in resistance training seemed to be quite high. The U.S. Consumer Product Safety Commission (CPSC) reported injury situations related to resistance training through its National Electronic Injury Surveillance System (NEISS). In 1979, NEISS reported approximately 17,000 injury incidents related to resistance training, while by 1987, this number had dropped to slightly over 8,000, including children and adolescents requiring emergency treatment (Faigenbaum et al., 1996). Additionally, the impact of resistance training on cartilage growth and soft tissue injuries in adolescents has always been a focus of concern. Due to the incomplete development of bones, adolescents have insufficient mechanical strength, thus increasing the risk of acute injuries (Lloyd et al., 2016; Nobari et al., 2024). This is also one of the reasons why resistance training was once considered potentially risky for adolescents. Studies have also shown that soft tissue injuries are more common in adolescents undergoing resistance training, especially in the hands and upper body regions (Jones et al., 2000; de Salles Painelli, 2023). One possible reason is that they often focus excessively on training the superficial muscles (such as the pectoral muscles and biceps), while neglecting the strengthening of core, back, and trunk muscle groups (Myers et al., 2017). Moreover, in the analyzed 97 cases, unsafe behaviors (63%), equipment malfunctions (37%), and lack of supervision (30%) were the main causes of injury (Jones et al., 2000). It is worrying that nearly half of the injuries occurred in the home environment, and four-fifths of the injuries were caused by improper use of home weight training equipment (Jones et al., 2000). Therefore, weight training or weight-bearing activities require professional supervision to prioritize and ensure safety and to minimize the risk of injury (Stricker et al., 2020).

4.2 Modern Consensus: Safety Guaranteed Through Scientific Training

Modern evidence indicates that appropriate resistance training for adolescents does not affect their health and has numerous benefits. Studies have shown that childhood and adolescence are critical periods for bone modeling and remodeling (Faienza et al., 2020). During this period, resistance training is crucial for supporting the growth and bone development of adolescents. (Malina, 2006; Tumkur Anil Kumar et al., 2021). Additionally, a review study on the safety of resistance training for adolescents revealed that 22 articles discussed the occurrence of sports injuries in adolescents during resistance training, and 10 of these reports described the related injury situations. However, in Malina's review, only 3 cases (including 2 cases of shoulder strain and 1 case of non-specific thigh pain) led to temporary interruption of training (Malina, 2006). Moreover, two studies on shoulder injuries further explored the possible subclinical musculoskeletal manifestations in early male adolescents during resistance training, and imaging examinations did not reveal any bone, epiphyseal plate, or muscle injuries (Tsolakis et al., 2004; Lillegard et al., 1997). Therefore, the research indicates that one of the main reasons for skeletal injuries in adolescents is the lack of necessary guidance and appropriate supervision (Malina, 2006).

Under the guidance of science and strict supervision, resistance training is a safe and effective intervention measure. The data from NEISS is based on patient self-reports, which attribute injuries directly to resistance training, but there may be inaccuracies (Faigenbaum et al., 1996). Additionally, after further analysis, these data cover different training forms, including resistance training, Olympic weightlifting, and powerlifting. Olympic weightlifting is a sport that emphasizes complex neural activation and explosive movements, aiming to push the body to its performance limit (Marino, 2022). In contrast, powerlifting includes movements such as 1RM squats and bench presses, which require a significant amount of force output and may lead to injuries (Syrett et al., 2022). The NEISS data from 2002 to 2005 indicated that approximately two-thirds of resistance training-related injuries among children aged 8 to 13 involved the hands and feet. Therefore, it can be concluded that most injuries suffered by adolescents during resistance training are accidental. If stricter safety guidelines are implemented and more professional supervision is provided, such accidents can be avoided (Myer et al., 2009).

In conclusion, a large amount of scientific evidence and clinical data indicate that when following appropriate training guidelines and procedures, resistance training is safe and effective for adolescents. It will not affect their physical development and can bring significant health benefits (Chaabene et al., 2020; Tumkur Anil Kumar et al., 2021).

5. Practical Guidelines and Implementation

As understanding of resistance training for adolescents has deepened, some professional organizations have issued guidelines in their medical and fitness position statements, clarifying the safety principles and practical requirements of resistance training for adolescents. In 2014, the United Kingdom Strength and Conditioning Association (UKSCA) released an official position statement on resistance training for adolescents, which was reviewed and endorsed by multiple authoritative professional organizations in the fields of sports medicine, sports science, and pediatrics (Lloyd et al., 2014). The UKSCA's statement emphasized technical, load, and ability issues in adolescent resistance training and was committed to ensuring the safety of adolescent resistance training by optimizing training variables. Additionally, other organizations have also issued similar statements based on comprehensive considerations of adolescent resistance training. The British Association of Sport and Exercise Sciences (BASES) released guidelines for adolescent resistance training in 2004, recommending that adolescents engage in at least two safe and effective resistance training sessions per week (Lloyd et al., 2012). The National Strength and Conditioning Association (NSCA) of the United States issued its first position statement on adolescent resistance training in 1985 and revised it in 1996 and 2004 (Faigenbaum et al., 1996; Faigenbaum, 2009). These position statements and guidelines not only reflect the importance of resistance training in adolescent growth and its widely recognized safety, but also provide clear bases for the scientific implementation of resistance training (Lloyd et al., 2012). For instance, these guidelines clearly define key elements such as warm-up activities, training methods, loads and intensities, sets, intervals, and training frequencies, providing a clear framework for constructing scientific, safe, and efficient training programs.

Warm-up activities are an integral and important part of resistance training programs for young individuals and their benefits have been widely studied and recognized. Firstly, warm-ups play a key role in preparing youth for resistance training and reducing the risk of injury (Afonso et al., 2024). In addition, it provides youth athletes with the necessary physiological and psychological conditioning to enhance training performance (Kyranoudis et al.,

2021; Till et al., 2021). In general, warm-up activities usually consist of low-intensity aerobic exercise as well as static and dynamic stretching (Faigenbaum et al., 2006; Ribeiro et al., 2021). In recent years, the effectiveness of static stretching as a traditional warm-up modality has been progressively questioned, and its limitations have been revealed by an increasing number of studies (Behm et al., 2021; Siebert et al., 2022). Meanwhile, dynamic stretching has gradually become the focus of attention in warm-up activities due to its significant advantages. Dynamic warm-up is not only effective in increasing body temperature, but also enhances the excitability and flexibility of motor units, while significantly improving dynamic motor performance (Faigenbaum et al., 2006; Sople & Wilcox, 2024). Related research has shown that adolescents who perform a moderate to high intensity dynamic warm-up prior to resistance training can significantly enhance strength performance (Ribeiro et al., 2021). Therefore, it is often recommended that adolescents spend 10 to 15 minutes warming up prior to resistance training to activate key muscle groups and increase joint mobility (Myburgh et al., 2020). A well-designed dynamic warm-up program provides a solid foundation for subsequent training and ensures a smoother and more efficient training process (as shown in Table 1).

Table 1: Implementation for Warm-up Before Resistance Training for Adolescent

Content category	Detailed description	Main Source
Advantages	-Provide physical and mental conditions; -Reduce the risk of injury; -Improve training performance; Enhance muscle group and joint activation	Kyranoudis et al. (2021); Till et al. (2021)
Type	Low-intensity aerobic exercise; Static stretching; Dynamic stretching	Faigenbaum et al. (2006); Ribeiro et al. (2021)
Recommended Time	10 to 15 minutes	Faigenbaum et al. (2009); Myburgh et al. (2020)
Summary	Before adolescents engage in resistance training, it is necessary to perform 10 to 15 minutes of warm-up exercises. The warm-up activities can include low-intensity aerobic exercises and stretching exercises. Static stretching has limitations and may temporarily reduce muscle strength and explosive power. Compared with static warm-up, dynamic warm-up has the advantages of increasing body temperature, enhancing neuromuscular excitability, flexibility, dynamic performance, and increasing power output. Therefore, it is recommended to choose a warm-up method of low-intensity aerobic exercise combined with dynamic stretching before strength training.	

Resistance training has been widely recognized as an effective method of improving adolescents' fitness through scientifically designed training modalities and variety. Although a wide range of apparatus and equipment is available for resistance training, practitioners must prioritize the adolescent's technical proficiency and foundational training level to ensure safety and effectiveness in all situations (Lloyd et al., 2014). The principles of resistance training suggest that the training modalities should focus on extensive and multi-joint muscle groups to maximize training effectiveness (Hammerová et al., 2020). Therefore, compared to single-joint muscle action, the training model should fully incorporate multiple muscle action types, particularly incorporating concentric (muscle shortening) and eccentric (muscle lengthening) phases of contraction into resistance training to optimize training effects and muscle adaptations (Burgos-Jara et al., 2023). For instance, different muscle actions contribute uniquely to training outcomes: concentric actions are particularly effective in developing strength and power, whereas eccentric actions are crucial for promoting muscle hypertrophy, enhancing neuromuscular control, and increasing flexibility (Kudiarasu et al., 2021; Liang et al., 2024).

In resistance training, the effectiveness of training is not only dependent on the training modality, but is also influenced by a combination of key parameters such as intensity and volume. Intensity is generally defined as the mechanical and metabolic stresses exerted during resistance training, whereas volume is defined as more repetitions in shorter rest intervals (Hurst et al., 2022). Resistance raining programs need to be developed with volumes and intensities selected appropriately in relation to individual goals. Safety should always be prioritized over athletic performance enhancement for adolescents or beginners (Brunsden, 2023). Therefore, the variables of volume and intensity must be balanced in resistance training to maximize physiological adaptations and reduce the risk of injury (Lloyd et al., 2014). Whilst high intensity and high volume resistance training can significantly stimulate muscular strength and muscle hypertrophy, they often lead to trainer fatigue, which in turn triggers distortions in movement form (Wilke & Mohr, 2020). Research has shown that adolescents exposed to excessive intensity (external loading), especially in technically immature individuals, are prone to acute injury (Pillitteri et al., 2024). In order to maximize muscle activation, improve motor unit recruitment, and reduce sport risk, adolescents should prioritize correct technique for each training movement under light volume or appropriate intensity conditions (Stricker et al., 2020).

Interval rest is a crucial variable when developing a youth resistance training program. Research has shown that rest between sets can help young individuals recover from fatigue more quickly (Behm et al., 2008; Staniszewski et al., 2024). A rest period between sets that is too short may negatively affect strength performance and power during training (Chang et al., 2022). Typically, the recommended interval time for resistance training in adults is 2-3 minutes (Longo et al., 2022). However, adolescents typically recover from fatigue faster due to greater flexibility and more excellent recovery of muscle tissue (Faigenbaum et al., 2019).

The training frequency is a critical variable often overlooked in the exercise prescription of youth resistance training. Current recommendations are that children and young individuals should perform resistance training two to three times per week. This recommendation is supported by research indicating that training intervals of 48 to 72 hours are sufficient to ensure adequate recovery in young individuals (Ramirez-Campillo et al., 2022). Furthermore, studies have shown that a minimum frequency of two resistance training sessions per week is necessary to promote significant increases in muscle mass (Brigatto et al., 2019). Given that youth are involved in multiple physical activities at school and are in a phase of growth and development, it is important to consider their physiological needs when designing training programs. Ensuring adequate rest and recovery is vital when developing a resistance training program. Based on the available evidence, practitioners can recommend that youth train twice weekly for approximately 60 minutes to enhance fitness and athletic performance (Santos et al., 2022).

Potential risks of exercise can be effectively reduced by identifying and adjusting key procedural variables when implementing resistance training programs for adolescents. In resistance training, it is recommended that adolescents start with low loads, small weights and few repetitions to gradually familiarize themselves with the movements and establish good training habits (dos Santos Duarte Junior et al., 2022). A moderate training pace is essential for beginners. Table 2 show the recommended variables for different training phases to support the effective implementation of a resistance training program (Faigenbaum et al., 2009). Research suggests that moderate fluctuations in speed should be maintained during training (Galiano et al., 2022). For example, using low loads and focusing on movement control exercises can help strengthen foundation and adaptation training in the early stages of training. In addition, properly designed training cycles can lead to greater training gains in the long term. By dynamically adjusting training programs, plateaus can be avoided during the 8–12-week training period, thereby maintaining a continuous increase in training effects (Hackett et al., 2018; Molinari et al., 2022; Kataoka et al., 2024).

Table 2. Key Variables in Resistance Training Programme Implementation

	Novice	Intermediate	Advanced
Muscle Action	ECC and CONc	ECC and CONc	ECC and CONc
Exercise Choice	SJ & MJ	SJ & MJ	SJ & MJ
Intensity	50 – 70% 1 RM	60 – 80% 1 RM	70 – 85% 1 RM

Volume	1 – 2 sets x 10 – 15 reps	2 – 3 sets x 8 – 12 reps	≥ 3 sets x 6 – 10 reps
Rest Intervals	1 min	1 – 2 min	2 – 3 min
Velocity	Moderate	Moderate	Moderate
Frequency	2 – 3 d/wk	2 – 3 d/wk	3 – 4 d/wk

Note: ECC = eccentric; CONc = concentric; SJ = single Joint; MJ = multi-joint; 1 RM = one repetition maximum. (Main Source: Quoted from Faigenbaum et al., 2009)

In addition to the empirical recommendations from various studies regarding effective variables in the implementation of resistance training for adolescents, relevant professional associations have also successively formulated and published normative guidelines for resistance training in this age group. Table 3 summarizes the recommendations and guidelines for resistance training programs for adolescents from authoritative bodies such as the UK Strength and Conditioning Association (UKSCA), the National Strength and Conditioning Association (NSCA), the American Academy of Pediatrics (AAP), the International Olympic Committee (IOC), and the Canadian Society for Exercise Physiology (Behm et al., 2008; Lloyd et al., 2012; Lloyd et al., 2014; Bergeron et al., 2015; Lloyd et al., 2016; Stricker et al., 2020). These programs and guidelines provide practitioners with scientific references to support adolescents in safely and effectively engaging in resistance training, thereby promoting the improvement of overall physical fitness and healthy development.

In summary, following the correct guidelines and procedures for training is an effective way to avoid sports injuries and improve fitness. Qualified instruction and supervision play a crucial role in adolescent resistance training, helping to improve training outcomes and provide multiple health benefits such as enhanced bone health and improved body composition (Stricker et al., 2020). Physical education teachers, coaches, and healthcare professionals should have a thorough understanding of the practical guidelines for resistance training to ensure that training programs are implemented in a scientific manner.

6. Future Directions

6.1 Standardization of Resistance Training Supervision

After three decades of development, society's perception of the benefits of resistance training for adolescents has shifted from the earlier misconception that it has "adverse effects" to a scientific consensus that it is "safe and effective". A large body of evidence indicates that standardized resistance training has multiple benefits for the physical and mental health of adolescents (McHenry & Nitka, 2022). Although its safety has been fully confirmed, the past controversy about its high risk of injury has been overturned by modern evidence-based research. However, the corresponding effective supervision has long lacked standardized definitions. Currently, although several professional institutions have issued guiding recommendations for resistance training for adolescents, there is still a lack of empirical research to support the question of "what standard supervision can minimize the risk of injury to the greatest extent" (Behm et al., 2008; Bergeron et al., 2015; Lloyd et al., 2016; Stricker et al., 2020). Existing literature mostly focuses on the training methods and effects for the teenage population, and there are few studies that systematically explore what structured and operational supervision processes sports teachers, parents, or coaches should specifically implement. However, establishing an effective supervision system is the core to ensure the safety and effectiveness of resistance training. Therefore, the key direction of future research lies in promoting the standardization and empirical validation of the supervision process. An empirical study highlighted the impact of different types of supervision on children's physical fitness, and the results showed that "supervision plus structured training" was the key factor to improve adolescents' physical fitness (Pinho et al.,2024). Therefore, by clarifying the qualifications, responsibilities, and specific intervention behaviors of the supervisors, and verifying their impact on the safety and effectiveness of training, a solid practical guarantee can be established for adolescents to participate in resistance training.

6.2 The Confounding Effect of Gender and Maturity

Although resistance training has been widely recognized for its health benefits in adolescents, the current research framework still has systematic omissions regarding the two key variables of gender and maturity. Currently, most empirical studies and professional guidelines only distinguish training loads based on training experience (such as

beginners and experts), without fully considering the possible differences in physiological traits and training responses among adolescents at different maturity stages or of different genders (Behm et al., 2008; Lloyd et al., 2012). There is evidence suggesting that significant differences between male and female adolescents in hormone levels, neuromuscular function, and the timing of puberty initiation may directly affect their strength development patterns, motor coordination abilities, and recovery efficiency after training (Rudroff et al., 2013). Ignoring the influence of these intrinsic biological factors makes it difficult to accurately assess the real effects of resistance training on the adolescent population. Therefore, future research should incorporate the gender and biological maturity of participants as core covariates into the analytical framework to control for potential confounding effects. A recent meta-analysis has shown that participation in resistance training in late adolescence in adolescent male athletes at various stages of maturity leads to more effective gains in muscle strength and explosive power (Lin et al., 2025). These findings provide empirical support for resistance training among adolescents at different maturity levels. Through this approach, it will be possible to more purely reveal the independent effect of resistance training itself, thereby providing a solid basis for establishing more individualized and scientific training guidelines.

6.3 Promotion and Application of Education

Although a large amount of current research evidence indicates that resistance training is safe and effective for adolescents, its application in school physical education curricula is still not satisfactory. In many countries, parents and students still believe that resistance training is detrimental to the physical development of adolescents and is even prone to causing sports injuries. The current physical education teaching syllabus still focuses mainly on competitive sports such as ball games and track and field, neglecting the importance of resistance training (An et al., 2022). It is worth noting that studies have shown that, under the premise of scientific supervision and reasonable load management, systematic resistance training has a lower risk of sports injuries compared to competitive sports (Lloyd et al., 2014). This indicates that there is a deviation in the public's perception of the risks of resistance training. In future physical education teaching practices, based on safety evidence, corresponding educational courses should be developed under effective supervision, and resistance training should be actively incorporated into the school physical education system. For example, the 2025 study by Vanaclocha-Amat et al., a research protocol, focused on the promotion and implementation of resistance training in schools (Vanaclocha-Amat et al., 2025). In addition, an interdisciplinary collaboration mechanism should be actively constructed to promote the deep integration of professional forces in the fields of sports science, education practice and medicine and health, so as to provide solid scientific support for it.

7. Conclusion

Overall, under the supervision and guidance of science, resistance training for adolescents is safe and effective. After a study period of over thirty years, the understanding of resistance training for adolescents has changed from being regarded as a risky activity in the 1980s to being widely recognized as having positive value in the 21st century. This consensus was established over a period of thirty years. Current empirical research has confirmed that under supervised and scientifically sound resistance training, the overall athletic performance of adolescents, including strength, explosive power, etc., as well as their physical and mental health levels, can be significantly improved. In addition, relevant professional associations in several countries have released guidance documents for resistance training for adolescents, providing scientific suggestions on training intensity, duration, and load, among other aspects. However, most of the existing studies have not clearly defined the specific criteria for "effective supervision", which may lead to insufficient actual supervision and thereby affect the safety of training. Given the particularity of the adolescent group, many studies also ignore the potential confounding effects of gender and maturity on the training outcome. Therefore, future research should focus on formulating standardized norms for effective supervision and deeply explore the influence mechanisms of the aforementioned confounding factors. Although current research results generally support the positive effects of resistance training for adolescents, there are still some doubts among parents and students. In the future, it is necessary to enhance the correct understanding of resistance training among adolescents and promote its scientific application in physical education. Based on relevant guidelines, it is important to develop targeted resistance training programs, promote the popularization of resistance training in school education, and ensure the safety and controllability of the training process. These are important directions for future practice.

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Appendix:

Table 3: UKSCA, NSCA, AAP, CSEF, IOC Resistance Training Guidelines

Qrganization	UKSCA	NSCA	AAP	CSFE	IOC
General guideline	- RT programme is age-related but not age-determined; - Qualified instruction; - Determination of participants' training status (stage of maturity, age at training, basic movement skills, technical proficiency, current strength level and psychological factors)	- Scientific instruction, effective supervision and a safe training environment; - Choose the right program and intensity of training to keep the exercise fresh and challenging; - Consider individual needs and concerns (e.g. weight, maturity level, health status); - Focus on proper exercise technique and coordination; - Finish with a cool down, including stretching.	- Correct technique and strict supervision; - Resistance training recommended after 7-8 years of age; - No anabolic steroids allowed.	- Qualified instruction and supervision; - Consideration of individual maturity and training experience; - A safe environment; -Learning the correct techniques.	- Considering differences in maturity and physical growth; - Implementing scientific programmes and responding to athletes' needs; - Work on psychological development; - Developing social skills and encouraging participation in sport;
Exercise session	- Use "child-sized" machines; - Free weight training over mechanical training; - Focus on muscle strength and basic athleticism; - Progressively more advanced exercises	- 5-10 minute dynamic warm-up; - Exercises the major muscle groups; - Multi-joint exercises before single-joint exercises; - Large muscle groups in front, small muscle groups in the back;	- Powerlifting or bodybuilding exercises are not recommended; - Proper warm-up and cool-down; - Use a full range of exercises to work the major muscle groups.	- 5-10 minute warm-up; - Strengthening upper body, lower body and midsection; - Balance and coordination exercises; - Low intensity and dynamic stretching to cool down; - Systematic changes to the training programme.	- Provide an enjoyable atmosphere; - Design for variety and variability in sport; - Optimise sports performance; - Athletes with sports injuries should not participate in training.

Intensity and volume	- Beginners: 1–2 sets of 8–12 exercises, 6–10 reps per set; bodyweight or 50–70% 1-RM; moderate–fast pace; 1 min rest; 2–3 sessions/week; 48–72 h recovery. - Intermediate: 2–4 sets of 6–10 exercises, 3–6 reps per set; 60–80% 1-RM; moderate–fast pace; 1–2 min rest; 2–3 sessions/week; 48–72 h recovery. - Experienced: 2–4 sets of 5–8 exercises, 3–6 reps per set; 70–85% 1-RM; moderate–fast pace; 1–2 min rest; 2–3 sessions/week; 48–72 h recovery.	- After starting with 1-2 sets ≤ 60% 1RM for familiarisation, 1-3 sets of 6-15 reps and around 80% 1RM for strength exercises. - After familiarisation, strength exercises 1-3 sets of 6-8 repetitions.	- No weights for beginners, gradually increase to body weight or other forms of resistance. - 2-3 sets, 8-15 reps, 2-3 times; 20-30 minutes; - RT sessions last a minimum of 8 weeks.	- Resistance training twice a week; - 8-12 sessions; - Training performed from a light load (60%).	-